Data Communication and Computer Networks

DCCN Question Bank | Unit 1 to 5 | Solved

All 44 long questions from the question bank, answered with diagrams, tables and worked numericals.

How to use this solution set

- Numbering follows your document exactly: **Qn here is question n in the question bank**, so you can work through them side by side.
- Every answer is written in the order it should be written in the exam - heading points first, then the expansion. The numbered points are the marking scheme.
- Answers are grounded in your lecture slides and in Forouzan, the prescribed text; where the same topic is asked twice in the bank (DNS, FTP, HDLC, multiplexing) the second answer takes a different angle rather than repeating the first.
- Numericals are worked line by line: the three subnetting questions, the CRC question and the Hamming example can be reproduced in the exam in under five minutes each.
- Diagrams are included wherever the question implies one - practise drawing them, because a labelled diagram carries marks of its own.

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Answers | 44 long questions

Numbering follows your question-bank document exactly, so answer n here is question n there. Each answer opens with the heading points and then expands them - write it in that order in the exam.

Q1. Explain the fundamentals of data communication and discuss the essential components required for effective communication in a network system.

Definition. Data communication is the exchange of data between two devices through some form of transmission medium such as a wire cable. *Telecommunication* means communication at a distance; *data* means information in whatever form the two parties agree upon.

Effectiveness rests on four fundamental characteristics:

- **Delivery** - data must reach the correct destination, and only the intended user.
- **Accuracy** - data must arrive unchanged; changed and uncorrected data are unusable.
- **Timeliness** - data must arrive in time, in the same order, and without delay for real-time audio and video.
- **Jitter** - the variation in packet arrival time must be small; large jitter shows up as uneven video.

Five essential components.

1. **Message** - the information to be communicated: text, numbers, pictures, audio, video.
2. **Sender** - the device that sends the message: computer, workstation, telephone handset, video camera.
3. **Receiver** - the device that receives it: computer, telephone handset, television.
4. **Transmission medium** - the physical path: twisted pair, coaxial cable, fiber-optic cable, radio waves.
5. **Protocol** - the set of rules governing the communication: what, when and how. Without it two devices may be connected but not communicating.

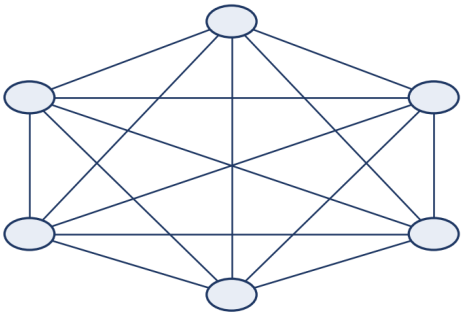
Data-flow modes complete the picture: **simplex** is unidirectional (keyboard, monitor), **half-duplex** allows both directions but not at once (walkie-talkie), and **full-duplex** allows both at once with the capacity divided between them (telephone).

Memory hook - D-A-T-J for the characteristics and M-S-R-M-P for the components.

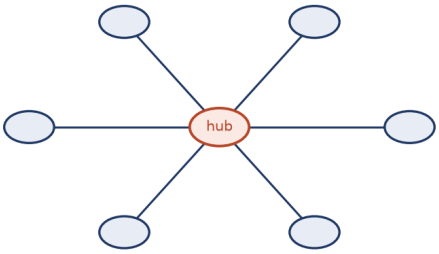
Q2. Describe different network topologies and explain how their structures influence communication and connectivity in computer networks.

Definition. The topology of a network is the geometric representation of the relationship of all the links and linking devices, called nodes, to one another.

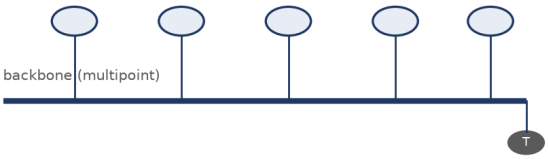
Mesh: n(n-1)/2 links, n(n-1) ports



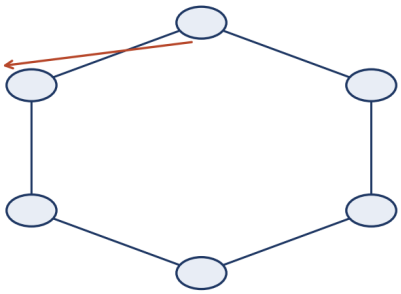
Star: n links, one per device to the hub



Bus: taps on a single backbone cable



Ring: n links, each device repeats the signal



Topology	Links for n devices	Advantages	Disadvantages
Mesh	n(n-1)/2, ports n(n-1)	Least traffic, robust, private, easy fault isolation	Huge cabling and port cost
Star	n	Robust per link, easy fault identification	Hub is a single point of failure
Bus	1 backbone plus drop lines	Least cabling	Hard to reconnect, reflections at taps, backbone failure kills all
Ring	n	Easy to add a device, repeater per node	Unidirectional traffic, a break disables the ring

How structure influences communication. Mesh gives every pair a dedicated path, so there is no contention and delay is low and predictable, at O(n squared) cost in cable and ports. Star centralises all traffic in the hub, so the hub becomes the bottleneck and the single point of failure, but each link runs at full rate. Bus and ring share one medium, so throughput falls as stations are added - bus because collisions rise, ring because access is serialised by the token. A **hybrid** combines them, for example a star backbone with bus segments, which is what most campus networks are.

Choosing: mesh for backbones and telephone regional offices, star for LANs, bus for the cheapest small network, ring where the repeater per node is needed for distance.

Q3. Explain the role of standardization in data communication and discuss its importance for ensuring interoperability between different networking devices.

- **Standard** - an agreed set of rules or requirements that provides guidelines to manufacturers, vendors, government agencies and service providers.
- **Role**: it fixes the format of frames, addresses, connectors, voltages and procedures, so equipment from different vendors can interoperate without a private agreement.
- **Market effect**: standards create and maintain an open and competitive market, because a customer is not locked into one supplier.
- **De facto** standards arise by convention or market dominance without formal approval (the IBM PC, early modem standards).
- **De jure** standards are recognised officially by a standards body (IEEE 802.3 Ethernet, ITU-T recommendations).

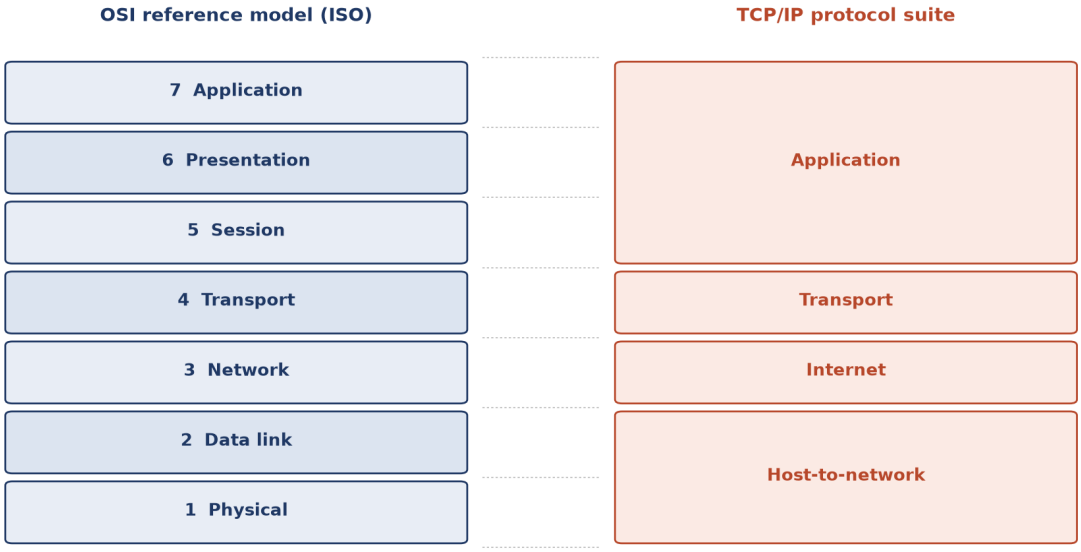
Organisations. ISO produced the OSI model; **ITU-T** sets telecommunication standards; **IEEE** runs Project 802 for LANs; **ANSI** is the US national body; **EIA** sets interface standards such as RS-232; the **IETF** publishes Internet standards as RFCs, which mature from proposed standard to draft standard to Internet standard.

Importance for interoperability. Layered standards let one layer change without disturbing the others, so a new physical medium can be introduced without rewriting the applications above it. The OSI and TCP/IP models are the clearest example: a web browser written to HTTP works over Ethernet, Wi-Fi or fiber because each layer's interface is standardised.

One-line answer: standards turn a collection of devices into a network - without them, every link would need a private agreement.

Q4. Discuss the OSI reference model and explain the functions of each layer in facilitating communication between systems.

Introduction. ISO produced the Open Systems Interconnection model in the late 1970s. It divides the whole task of communication into seven ordered layers, each using the services of the layer below and offering services to the layer above.



One TCP/IP layer may cover several OSI layers; the two models describe the same work at different granularity.

1. **Physical** - physical transmission of bits: mechanical and electrical specifications of the interface and medium, representation of bits, data rate, bit synchronisation, line configuration, topology, transmission mode.
2. **Data link** - moves frames from one node to the next: framing, physical (MAC) addressing, flow control, error control, access control. It makes the physical layer appear error-free to the network layer.
3. **Network** - source-to-destination delivery of a packet across networks: logical (IP) addressing and routing.
4. **Transport** - process-to-process delivery of the entire message: port addressing, segmentation and reassembly, connection control, end-to-end flow and error control.

5. **Session** - establishes, manages and terminates sessions: dialog control and synchronisation by checkpoints.
6. **Presentation** - syntax and semantics: translation, encryption, compression.
7. **Application** - services to the user: file transfer and access management, mail services, directory services.

Mnemonic (7 to 1) - **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing.

Q5. Explain the TCP/IP reference model and compare its architecture with the OSI model in terms of functionality and practical implementation.

1. **Host-to-network** (OSI physical + data link) - transmits data between two devices on the same network.
2. **Internet** (OSI network) - host-to-host delivery using **IP** (datagrams, best effort), **ARP** (IP to physical), **RARP** (physical to IP), **ICMP** (errors and queries) and **IGMP** (group transmission).
3. **Transport** (OSI transport) - process-to-process delivery using **TCP** (connection-oriented, reliable) and **UDP** (connectionless, unreliable, fast).
4. **Application** (OSI session + presentation + application) - HTTP, SMTP, FTP, DNS, Telnet.

Point	OSI	TCP/IP
Layers	7	4 (5 in the Forouzan treatment)
Order of definition	Model first, protocols later	Protocols first, model describes them
Nature	General reference model, never fully implemented	Practical suite, deployed on the Internet
Session and presentation	Separate layers	Merged into the application layer
Network layer service	Connection-oriented and connectionless	Connectionless only; reliability moved to transport
Practical implementation	Used as a teaching and design guide	Used in every real host and router

Functionality comparison. Both are layered and both provide addressing, routing and reliable delivery; the difference is granularity. OSI separates session and presentation so their duties are visible and can be standardised independently, while TCP/IP leaves them to the application, which is why encryption (TLS) and character translation sit inside applications in practice. OSI's network layer promises both connection-oriented and connectionless service; TCP/IP keeps the internet layer connectionless and puts reliability in TCP, which keeps routers simple and fast.

Q6. Describe analog and digital signals used in the physical layer and explain their significance in data transmission systems.

- **Analog data** are continuous; **digital data** take discrete states. An **analog signal** has infinitely many levels of intensity over time; a **digital signal** has only a limited number of defined values.
- Data communication commonly uses **periodic analog signals** and **non-periodic digital signals**.
- A sine wave is described by **amplitude** (height, in volts or watts), **frequency** (cycles per second, Hz) and **phase** (position relative to time zero).
- **f = 1/T**, so 60 Hz has a period of 16.67 ms; **wavelength = propagation speed / frequency**, so 1 MHz in a cable at 2 x 10^8 m/s has a wavelength of 200 m.
- A **composite signal** is a sum of sine waves (Fourier analysis); its **bandwidth** is the difference between the highest and lowest frequency it contains.
- A digital signal with L levels carries log2 L bits per signal element, so more levels mean more bits per symbol but less noise margin.

Significance. The analog signal is what the medium actually carries, so every digital transmission must first be turned into a waveform (line coding) or onto a carrier (modulation). The bandwidth of the signal decides the bandwidth the medium must provide; the frequency decides the antenna size; and the number of levels decides the trade-off between bit rate and error rate. Analog transmission suffers cumulative noise, while a digital signal can be regenerated, which is why modern systems digitise early.

Exam line: analog is a continuum, digital is a set of steps - and every transmission problem is about fitting the steps into the bandwidth and noise of the medium.

Q7. Explain the concept of data rate limits and discuss the factors that influence the maximum transmission capacity of a communication channel.

- **Data rate limit** is the maximum number of bits per second that a channel can carry.
- It depends on three factors: the **bandwidth available**, the **number of signal levels** used, and the **quality of the channel** - that is, the level of noise.
- **Noiseless channel (Nyquist)**: bit rate = 2 x B x log2 L. The 2 comes from the sampling theorem - 2B samples per second, each carrying log2 L bits.
- **Noisy channel (Shannon)**: C = B log2 (1 + SNR), the theoretical maximum error-free rate.
- **SNR in dB** = 10 log10 SNR, so 30 dB means a power ratio of 1000.

1. Nyquist example: B = 6000 Hz with 4 levels gives 2 x 6000 x 2 = 24 kbps.
2. Shannon example: a telephone line with B = 3000 Hz and SNR = 3162 gives 3000 x log2 3163, about 34.88 kbps - the origin of the 33.6 kbps modem.
3. Combined: for 265 kbps over a noiseless 20 kHz channel, log2 L = 6.625, so L must be about 98.7 - use 128 levels (280 kbps) or accept 64 levels (240 kbps).
4. A channel with B = 10 MHz and SNR = 60 dB gives C = 10^7 x 19.93, about 199.3 Mbps.

Interpretation: Shannon says whether the target is physically possible; Nyquist says what modulation is needed to reach it. Noise, not bandwidth, is usually the binding constraint on copper.

Q8. Describe transmission impairment and explain how noise, attenuation, and distortion affect communication quality in networks.

Definition. Media are not perfect, so the signal at the end of the medium is not the signal that entered it. The three causes are attenuation, distortion and noise.

- **Attenuation** - loss of energy overcoming the resistance of the medium, so amplitude falls with distance. dB = 10 log10 (P2/P1); a negative value is a loss. Cable loss is quoted per kilometre, for example -0.35 dB/km. Remedy: amplifiers or repeaters, placed so the signal never falls into the noise.
- **Distortion** - change of shape. Each frequency component travels at its own speed, so components arrive with different delays and phases and their sum is no longer the original shape. It affects composite signals only. Remedy: equalisation, or a medium with a flat delay characteristic.
- **Noise** - unwanted signal added on the way: **thermal** (random electron motion, present in every device), **induced** (motors and appliances, and crosstalk from a neighbouring pair) and **impulse** (lightning, power-line spikes). Remedy: shielding, twisting, and a higher signal power.

Effect on quality. Attenuation reduces the signal-to-noise ratio at the far end; distortion makes the eye pattern close so bits are misread even when the amplitude is adequate; noise sets the floor below which no signal can be distinguished. All three end in the same currency - bit error rate - which is why SNR is the number a link budget is built around.

Memory hook - A-D-N: **A**ttenuation loses it, **D**istortion bends it, **N**oise adds to it.

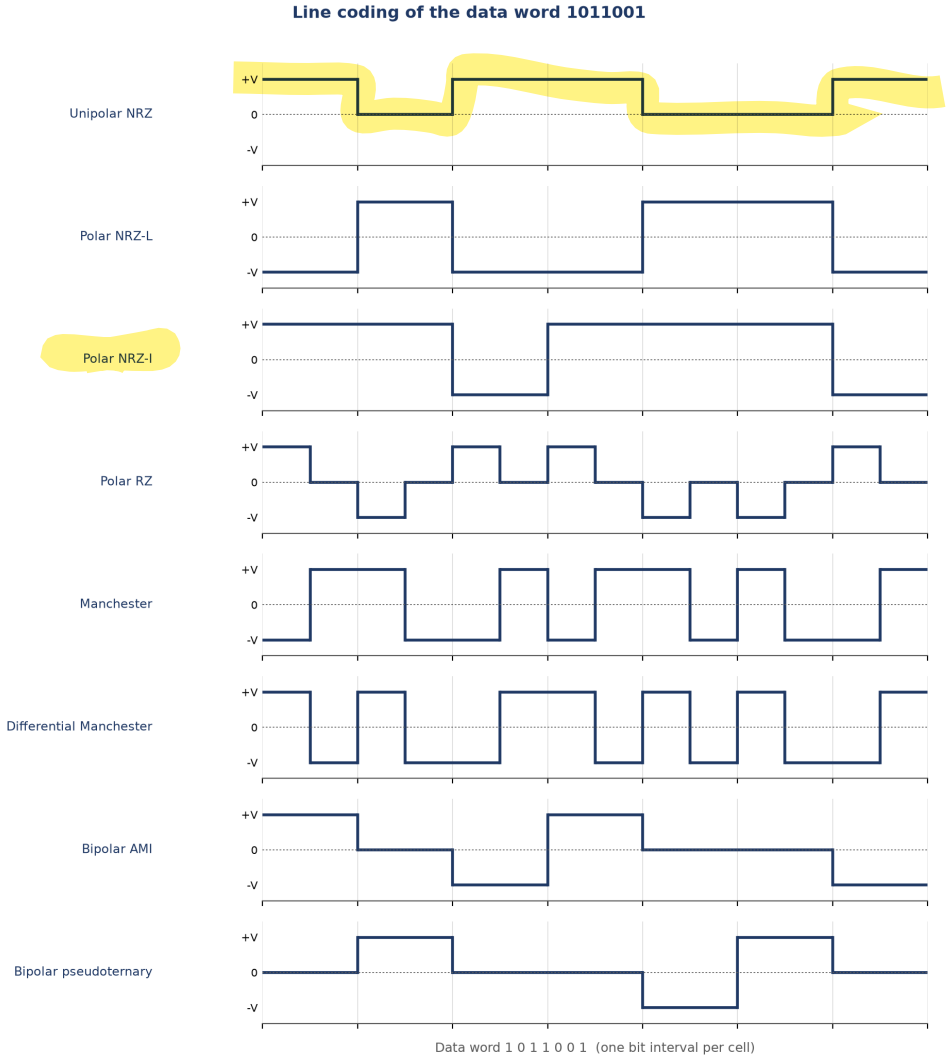
Q9. Explain the concept of digital transmission and discuss the role of line coding and sampling in transmitting digital data effectively.

Digital transmission sends data as a sequence of discrete signal levels. It is preferred because the signal can be regenerated at each repeater, errors can be detected and corrected, and voice, video and data can share one format.

- **Line coding** converts a sequence of bits into a digital signal, and the receiver decodes it back. It decides the DC content, the synchronisation and the bandwidth.
- **Unipolar NRZ**: 1 = +V, 0 = 0 V - simple, but has a DC component and no synchronisation.
- **Polar NRZ-L / NRZ-I**: levels on both sides of the axis; NRZ-I inverts on 1, so long runs of 1s are safe but long runs of 0s are not.
- **Polar RZ, Manchester and differential Manchester**: self-clocking, DC-free, but they need twice the bandwidth because each bit has two transitions. Manchester is used in 10 Mbps Ethernet.
- **Bipolar AMI and pseudoternary**: three levels, DC-free, with alternation that allows some error detection; a long run of the zero symbol loses synchronisation.

- **MLT-3**: three levels cycling 0, +V, 0, -V; changes only on 1s, so it needs about one third of NRZ's bandwidth - used in 100BASE-TX.

Sampling is the first step of analog-to-digital conversion: the signal is measured every Ts seconds, and by the sampling theorem fs must be at least 2 fmax, otherwise the signal is aliased. The three methods are ideal (impulses), natural (pulse top follows the signal) and flat-top (sample and hold, the practical one). Sampling is followed by quantization and encoding, giving the bit rate fs x n - for telephony, 8000 x 8 = 64 kbps.



Trade-off to state: synchronisation and DC-freedom cost bandwidth. That single sentence answers most comparison questions on line codes.

Q10. Describe different transmission modes in data communication and explain their significance in efficient data exchange between devices.

Transmission modes are classified in two independent ways: by **direction** of data flow, and by **how the bits are sent** across the link.

- **Simplex** - unidirectional; the whole channel capacity goes one way. Keyboard to computer, traditional monitor.
- **Half-duplex** - both ways but not at the same time; the transmitting device takes the entire capacity. Walkie-talkie.
- **Full-duplex** - both ways at once; the capacity is divided between the two directions. Telephone network.
- **Parallel transmission** - n wires carry n bits on one clock tick: fast, but costly and limited in distance by skew. Computer to printer.
- **Serial transmission** - one bit after another on one channel: cheap, suits long distance, and is run at a much higher clock. Modems, LANs.
- **Asynchronous serial** - each byte framed by a start bit (0) and one or more stop bits (1), gaps allowed; about 20 per cent overhead.

- **Synchronous serial** - no start or stop bits and no gaps; the receiver groups the bits, and many bytes are sent in a frame with a start and end byte.

Significance. Direction decides whether a protocol needs turn-taking and how much capacity each side gets; the parallel or serial choice decides cost against speed and the maximum distance; and the asynchronous or synchronous choice decides overhead against the need for a shared clock. Choosing wrongly shows up either as wasted bandwidth or as an unreliable link.

Rule of thumb: short and fast means parallel; long and cheap means serial; cheap and simple means asynchronous; efficient means synchronous.

Q11. Explain the concept of analog transmission and discuss how digital data and analog signals can be modulated for effective communication.

Analog transmission sends information by varying one characteristic of a continuous carrier, C(t) = A cos(2 pi fc t + phi). It is used whenever the medium is bandpass rather than low-pass - telephone local loop, radio, microwave, satellite.

Digital data on an analog carrier (digital-to-analog conversion).

- **ASK** - amplitude switched: 1 sends the carrier, 0 sends zero. Generated by multiplying unipolar NRZ with the carrier; bandwidth B = (1 + d)S.
- **FSK** - frequency switched: f1 = fc + delta-f for 1, f2 = fc - delta-f for 0; generated with NRZ and a VCO. Widest bandwidth, simplest hardware.
- **PSK** - phase switched: 0 and 180 degrees in BPSK; generated by multiplying polar NRZ with the carrier. Best noise immunity at ASK's bandwidth.
- **QAM** - amplitude and phase varied together, so L x M points carry log2(L x M) bits per symbol; used in modems and digital TV.

Analog data on an analog carrier (analog-to-analog conversion). AM varies the amplitude and needs 2B; **FM** varies the frequency and needs 2(1 + beta)B; **PM** varies the phase with the same bandwidth form as FM. FM and PM keep a constant envelope, so amplitude noise can be clipped away.

Why modulate at all. The carrier shifts the signal into the passband of the medium, makes antennas practical (a baseband signal would need an antenna kilometres long), allows many users to share one medium on different carriers, and lets a weak signal be repeated at a new frequency.

Q12. Explain the modulation of analog signals and discuss how modulation techniques support long-distance communication.

- **AM** - the amplitude of the carrier follows the modulating signal, so the modulating signal becomes the **envelope** of the carrier. Bandwidth = 2B. Simple and cheap, but noise adds to the amplitude and goes straight into the data.
- **FM** - the frequency of the carrier follows the modulating signal; amplitude stays constant. Bandwidth = 2(1 + beta)B. Rejects amplitude noise, which is why FM broadcast sounds cleaner than AM.
- **PM** - the phase of the carrier follows the modulating signal; the phase change appears as a frequency change proportional to the derivative of the modulating signal. Bandwidth as for FM.

How modulation supports long distance.

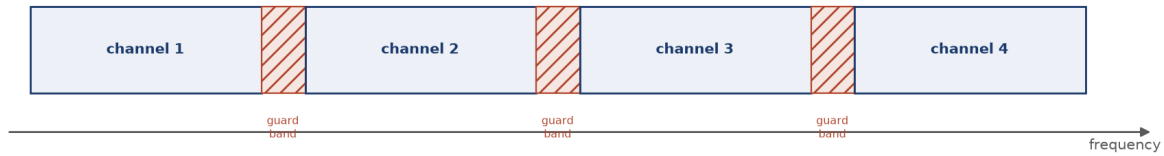
1. **Frequency translation** - a low-frequency voice signal cannot radiate efficiently, so it is moved onto a high carrier that propagates through the medium and can be beamed with a directional antenna.
2. **Antenna size** - the antenna must be about a quarter wavelength, so at 1 MHz it is about 75 m whereas at 5 kHz it would be kilometres long.
3. **Noise immunity** - FM and PM put the information in frequency or phase, so a limiter removes the accumulated amplitude noise; this is what lets a weak long-distance signal still be read.
4. **Multiplexing** - different carriers let many conversations share one long-haul medium (FDM), which is what made the first transcontinental telephone systems economic.
5. **Repeaters** - a repeater demodulates, regenerates and remodulates, so noise is not amplified along with the signal.

Bandwidth ladder to remember: AM = 2B, FM and PM = 2(1 + beta)B - so FM buys its noise immunity with (1 + beta) times the bandwidth.

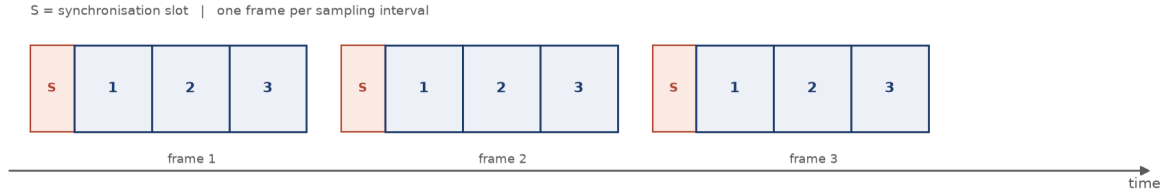
Q13. Describe the concept of multiplexing and explain the working principles of Frequency Division Multiplexing (FDM), Wavelength Division Multiplexing (WDM), and Time Division Multiplexing (TDM).

Definition. Multiplexing allows several signals to be carried over one link at the same time. The multiplexer combines n streams into one (many-to-one) and the demultiplexer separates them again (one-to-many). It is used whenever the link bandwidth exceeds what one signal needs.

FDM - each channel gets a slice of the frequency band



TDM - each channel gets a slot in every frame



1. **FDM** - an analog technique. The bandwidth of the link is divided into bands, one per signal, each signal is modulated onto its own carrier, and guard bands keep the channels apart. All signals travel at once. Example: AM and FM broadcast, cable TV, first-generation telephone carriers. Bandwidth needed = n x B + (n-1) x guard band.
2. **WDM** - FDM at optical frequencies. Several light signals of different wavelengths share one fiber, combined and split by a prism or diffraction grating. It is what makes terabit backbones possible.
3. **TDM** - a digital technique. Time is divided into frames and each frame into slots, one per channel; every channel uses the whole bandwidth but takes turns. Total rate = channels x bits per sample x sampling rate, so 32 x 8 x 8000 = 2.048 Mbps.
4. **Synchronous TDM** gives every stream a fixed slot whether or not it has data; **statistical (asynchronous) TDM** allocates slots only to active streams, which is efficient but needs an address per slot and buffers at both ends.

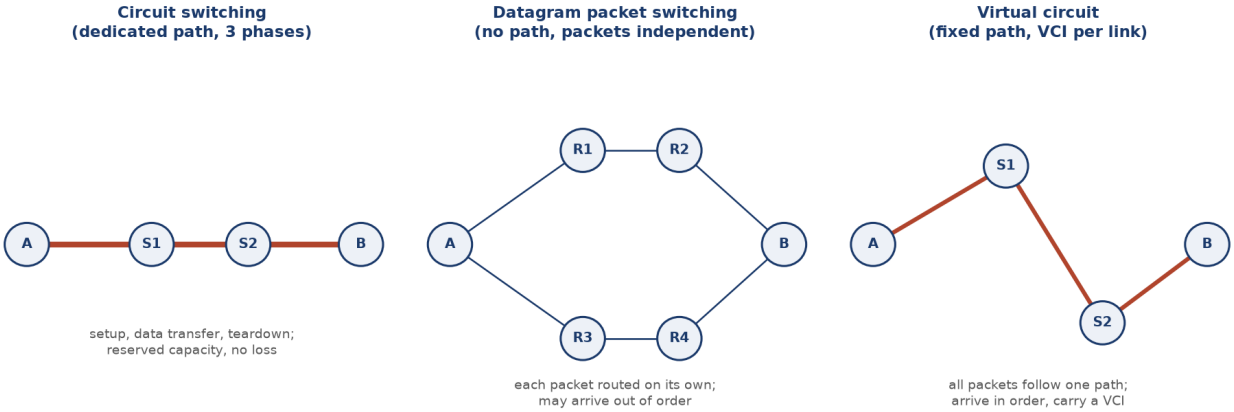
One-liner: FDM divides the spectrum, TDM divides the clock, WDM divides the colour. The **inverse multiplexer** does the opposite - it spreads one high-rate stream over several low-rate links.

Q14. Describe guided and unguided transmission media and explain their roles in data communication systems.

- **Guided media** provide a physical conduit: twisted pair, coaxial cable and fiber-optic cable. The signal is confined, so the link is secure and free from external interference (fiber completely).
- **Twisted pair** - two insulated copper wires twisted in a helix; the twists make induced noise equal and opposite on the two wires, so it cancels. UTP and STP forms; Category 5e and 6 for Ethernet.
- **Coaxial cable** - copper core, dielectric, braided shield and jacket. The shield gives better immunity and higher bandwidth than twisted pair; used for cable TV and broadband.
- **Fiber-optic cable** - glass or plastic core with cladding, carrying light by total internal reflection. Tbps bandwidth, very low attenuation, immune to EMI and crosstalk, and the hardest to tap.
- **Unguided media** carry electromagnetic waves without a conductor: **radio waves** (omni-directional, penetrate walls, multicast - radio and TV), **microwaves** (directional, line of sight, cannot penetrate walls - cellular, satellite, WLAN) and **infrared** (short range, closed area, line of sight - remote controls).

Role in a network. Guided media form the backbone and the local loop, where capacity and security matter most; unguided media provide mobility and reach places no cable can go - a ship, a village, a moving vehicle. Real networks combine them: fiber between cities, copper or Wi-Fi in the last hundred metres.

Q15. Explain the concept of circuit switching and discuss how it differs from packet switching in communication networks.



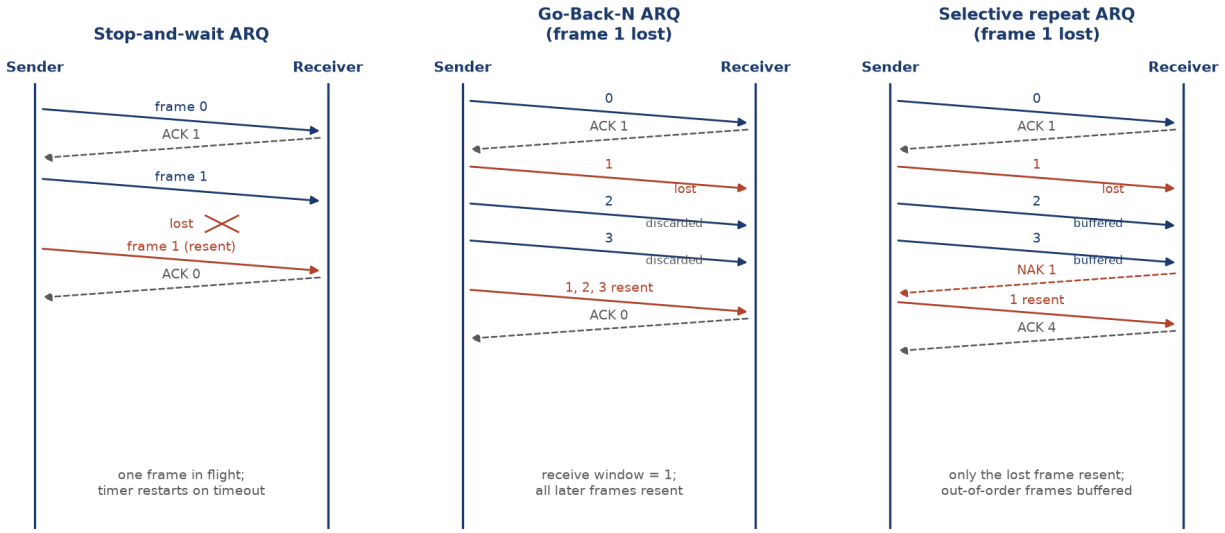
1. **Circuit switching** works at the physical layer. A path is reserved before transmission and held for the whole conversation; each link is divided into channels by FDM or TDM; data flow continuously and are not packetised. Three phases: setup, data transfer, teardown.
2. **Packet switching** works at the network layer (datagram) or data link layer (virtual circuit). The message is divided into packets, each with a destination address; no resource is reserved and packets are served first-come-first-served.
3. **Virtual circuit** is the middle case: a path is set up before transfer so packets arrive in order, but no capacity is reserved, and each packet carries a VCI instead of the full address.

Point	Circuit switching	Packet switching
Path	Dedicated and reserved	None (datagram) or fixed (virtual circuit)
Resource use	Wasted while the circuit is silent	Shared and efficient
Delay	Setup delay, then constant	Queueing delay per packet, variable
Order and loss	Order guaranteed, no loss	May reorder or drop
Suitability	Voice, constant-rate traffic	Bursty data, the Internet
Example	Telephone network	The Internet

One-line answer: circuit switching reserves the road and packet switching shares it - so the first guarantees delay and wastes capacity, the second does the opposite.

Q16. Explain the Go-Back-N ARQ protocol in detail. Discuss its working mechanism, advantages, and limitations with examples.

1. Sequence numbers are modulo 2^m. The **send window** holds up to 2^m - 1 frames; the **receive window** is 1.
2. The sender may transmit up to the window limit without waiting, and runs one timer for the **oldest** outstanding frame.
3. The receiver accepts only the frame with the expected number and returns a **cumulative ACK**; out-of-order frames are discarded.
4. If the timer expires, the sender resends that frame **and every frame after it** - hence the name.



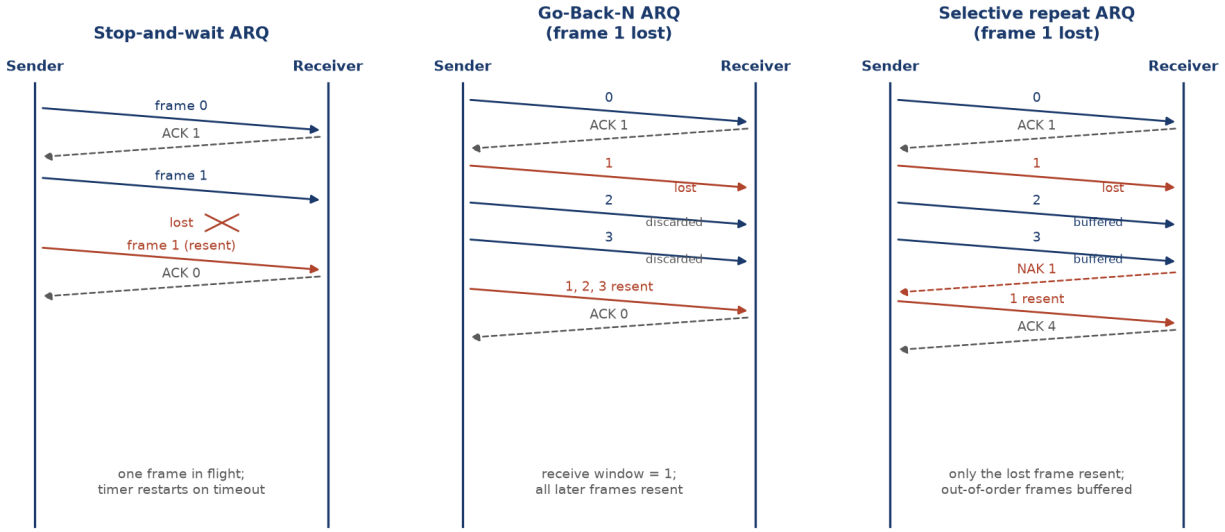
Worked example. With m = 2 the window is 3. Frames 0, 1, 2 and 3 are sent; frame 1 is lost. The receiver accepts 0, then discards 2 and 3 as out of order and keeps acknowledging 1. When the timer of frame 1 expires the sender resends 1, 2 and 3 - three frames for one loss.

- **Advantages**: many frames in flight, so utilisation is far better than stop-and-wait; one cumulative ACK can cover several frames, so a lost ACK is often harmless; one timer serves the whole window; the receiver needs no buffer.
- **Limitations**: on a lossy link it resends frames that arrived correctly, wasting bandwidth; the whole window is delayed by one loss; the receive window of 1 means it cannot reorder.

When to use: a link with low error rate and a long propagation delay, where the window matters more than the occasional wasted retransmission.

Q17. Explain the Selective Repeat ARQ protocol in detail. Discuss its working mechanism, advantages, and limitations with examples.

1. Send and receive windows are each at most **half of 2^m**, so a new frame can never be confused with a retransmission.
2. The receiver **buffers** out-of-order frames and marks them; they are delivered to the network layer only when the missing frame arrives.
3. The sender keeps a **timer per frame** and resends only the frame whose timer expires; a NAK can request one frame immediately.
4. ACKs are sent when data are delivered to the network layer, so one ACK may cover several frames.



Worked example. Frames 0 to 3 are sent; frame 1 is lost. The receiver buffers 2 and 3 and sends NAK 1. The sender resends only frame 1; when it arrives, frames 1, 2 and 3 are delivered together and one ACK covers all three. One retransmission instead of three.

- **Advantages**: only the lost frame is retransmitted, so bandwidth is not wasted; throughput stays high on error-prone links; the NAK gives faster recovery than a timeout.
- **Limitations**: the receiver must buffer, the sender needs a timer per frame, and the protocol is more complex to implement; the window is halved, so the maximum in-flight data are less than in Go-Back-N for the same m.

Compare in one line: Go-Back-N resends the window, selective repeat resends the frame.

Q18. Explain HDLC (High-Level Data Link Control) protocol in detail. Discuss its frame format and applications in communication systems.

- **HDLC** is a **bit-oriented** data link protocol for point-to-point and multipoint links. It provides framing, flow control and error control through ARQ, and transparency through bit stuffing.
- **Flag** (8 bits, 01111110) - delimits the frame and synchronises the receiver.
- **Address** (8 or more bits) - identifies the secondary station.
- **Control** (8 or 16 bits) - sequence numbers, ACK and frame type (I, S, U).
- **Information** (variable) - the user data from the network layer.
- **FCS** (16 or 32 bits) - the CRC for error detection.
- **Transfer modes**: NRM (unbalanced, secondary polled), ARM (unbalanced, secondary may initiate) and ABM (balanced, point-to-point).

Applications. HDLC is the basis of the ISO data link standards and of many proprietary protocols: it is used on leased lines between routers, in X.25 at the link layer (as LAPB), in frame relay (as LAPF), in ISDN (as LAPD) and in Cisco's serial encapsulation. Its frame structure survives in every bit-oriented link protocol because the flag-plus-FCS design is both simple and robust.

Contrast with PPP: PPP is byte-oriented, point-to-point only, and adds authentication and protocol multiplexing, but it reuses HDLC's frame layout.

Q19. Describe random access protocols in detail. Explain ALOHA and CSMA/CD with their advantages and limitations.

Random access means no station is superior and none is assigned control; each station with data decides for itself whether to send, using a procedure defined by the protocol. Collisions are possible and must be handled.

1. **Pure ALOHA**: transmit whenever you have a frame, with no sensing. A frame survives only if no other frame starts within one frame time before or after it, so the vulnerable period is two frame times. Throughput S = G e^(-2G), maximum 18.4 per cent at G = 0.5.
2. **Slotted ALOHA**: transmission only at slot boundaries, so the vulnerable period halves. S = G e^(-G), maximum 36.8 per cent at G = 1.
3. **CSMA**: sense the channel before transmitting. Persistence methods: 1-persistent transmits at once when idle, non-persistent waits a random time when busy, p-persistent transmits with probability p in the next slot.
4. **CSMA/CD**: sense before transmitting and **keep sensing while transmitting**. On collision, send a jam signal, abort, wait a random backoff time (binary exponential) and retransmit. Used in traditional Ethernet; requires the frame to be long enough for the collision to return before the sender finishes.

- **ALOHA advantages**: extremely simple, no coordination, works with any number of stations. **Limitations**: at most 18.4 per cent (36.8 per cent slotted) of capacity, and delay rises sharply with load.
- **CSMA/CD advantages**: much higher utilisation than ALOHA, and a collision is abandoned at once instead of wasting the whole frame. **Limitations**: collision detection needs to hear the medium while transmitting, so it cannot be used on radio; performance degrades at high load; and the minimum frame size ties the protocol to the cable length.

Numbers to quote: 18.4 per cent pure ALOHA, 36.8 per cent slotted, and CSMA/CD well above both at light load.

Q20. For a given IP address 10.0.0.0, the network has been divided into 32 sub-networks. Determine (i) the network ID, (ii) the default subnet mask, (iii) the new subnet mask, (iv) the number of IP addresses per subnet, (v) the usable hosts per subnet, (vi) the broadcast address of each subnet and (vii) the range of addresses in each subnet.

1. **Network ID**: 10.0.0.0. The first byte 10 lies in 1-126, so this is a **class A** address.
2. **Default subnet mask**: 255.0.0.0, that is /8 - one byte of net-id and three bytes of host-id.
3. **Subnet bits**: 32 subnets need log2 32 = **5 bits**.
4. **New subnet mask**: /8 + 5 = **/13**, written 255.248.0.0 (248 = 11111000).
5. **IP addresses per subnet**: 2^(32-13) = 2^19 = **524,288**.
6. **Usable hosts per subnet**: 524,288 - 2 = **524,286**.
7. **Block size** in the second octet = 2^3 = **8**, so the subnets step 10.0.0.0, 10.8.0.0, 10.16.0.0 ... 10.248.0.0.

Subnet	Network ID	Broadcast	Usable range
1	10.0.0.0	10.7.255.255	10.0.0.1 - 10.7.255.254
2	10.8.0.0	10.15.255.255	10.8.0.1 - 10.15.255.254
3	10.16.0.0	10.23.255.255	10.16.0.1 - 10.23.255.254
...	...	...	...
32	10.248.0.0	10.255.255.255	10.248.0.1 - 10.255.255.254

Routine: class, default mask, subnet bits = log2(subnets), new prefix = default + subnet bits, addresses = 2^host bits, usable = minus 2, block size = 2^host bits in the last subnet octet.

Q21. Explain IPv6 in detail. Compare IPv4 and IPv6 in terms of addressing scheme, header format, advantages, and limitations.

- **Addressing**: IPv6 uses **128-bit** addresses written as eight groups of four hexadecimal digits separated by colons; the address space is 3.4 x 10^38. Leading zeros in a group and one run of zero groups can be compressed (::).
- **Address types**: unicast, multicast and anycast - there is no broadcast.
- **Header**: a fixed **40-byte** base header (version, traffic class, flow label, payload length, next header, hop limit, source and destination address) followed by extension headers, which is why routers process it faster.
- **Other features**: IPsec built in, stateless autoconfiguration, no fragmentation by routers (only the source fragments), and no need for NAT.

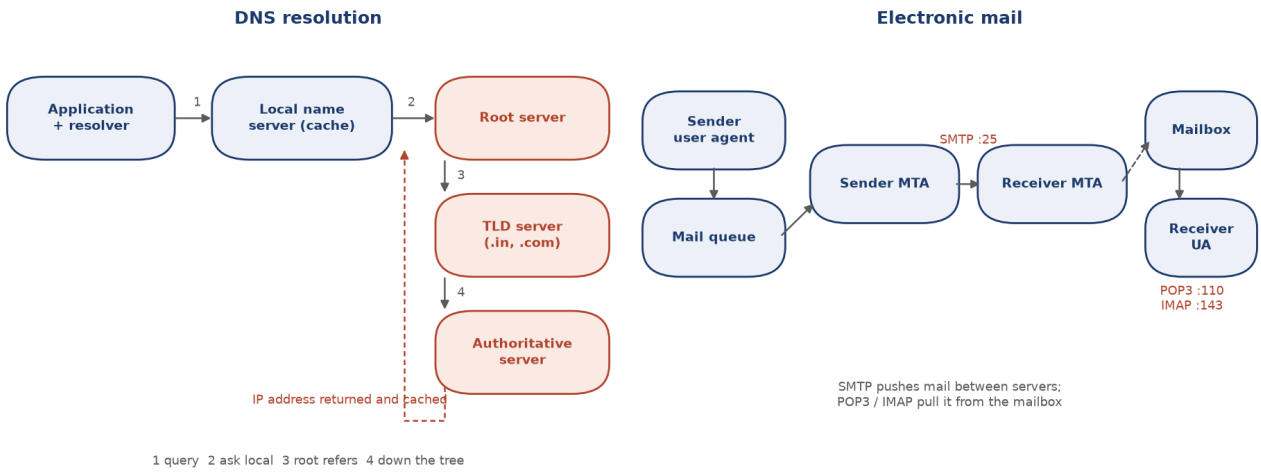
Point	IPv4	IPv6
Address size	32 bits, dotted decimal	128 bits, colon hexadecimal
Header	Variable, 20-60 bytes, 13 fields	Fixed 40 bytes plus extension headers
Checksum	Header checksum present	Removed - left to upper layers
Fragmentation	Routers and hosts	Source only
Security	IPsec optional add-on	IPsec part of the standard
Configuration	Manual or DHCP; NAT common	Stateless autoconfiguration; no NAT
Broadcast	Supported	Replaced by multicast and anycast

- **Advantages of IPv6**: enough addresses for any foreseeable growth; simpler, faster forwarding; built-in security and mobility; true end-to-end connectivity restored.
- **Limitations of IPv6**: IPv4 and IPv6 are not interoperable, so dual stacks or translation are needed during migration; the 128-bit addresses are harder to remember; and legacy equipment and applications still speak IPv4 only, which is why deployment has taken decades.

Exam line: IPv6 is not IPv4 with bigger numbers - the header was redesigned so that routers do less work per packet.

Q22. Explain the Domain Name System (DNS). Discuss its architecture, working mechanism, and importance in translating domain names into IP addresses.

- **Definition**: DNS is a hierarchical, distributed naming system that translates human-readable domain names into IP addresses, and back.
- **Architecture**: a tree of domains with the root at the top, served by a hierarchy of name servers - root, top-level domain, authoritative, and the local caching server each ISP runs. It works on the client-server model and uses **UDP port 53**.
- **Name space**: generic domains (.com, .org, .edu), country domains (.in, .us, .uk) and the inverse domain (in-addr.arpa) for reverse lookup.
- **Records**: A (IPv4 address), AAAA (IPv6), MX (mail server), CNAME (alias), NS (name server), PTR (reverse lookup).



1. The application asks the **resolver** to resolve a name.
2. The resolver checks its cache; on a miss it queries the **local name server**.
3. The local server checks its cache and zone file, then asks the **root server**, which refers it to the top-level domain server.
4. The TLD server refers it to the **authoritative server** for the domain, which returns the IP address.
5. The answer travels back and is cached at every step for its TTL; the resolver returns the address to the application.

- **Importance**: users cannot remember numeric addresses, and addresses change when a host moves or a provider changes. DNS decouples the name from the address, so a website keeps its name while the address behind it changes.
- It is also the mechanism that lets mail find the right server (MX records) and lets services be split across machines (CNAME and load balancing).
- Because it is distributed and cached, no single server holds the whole database, and the load of billions of queries a day is spread across the hierarchy.

One-line answer: DNS is the Internet's phone book - distributed, hierarchical, and cached, which is why a lookup usually takes milliseconds.

Q23. Explain File Transfer Protocol (FTP) and discuss its properties.

- **FTP** transfers files from one host to another over TCP and hides the differences between the two file systems, text conventions and directory structures.
- **Two connections**: a **control connection** on port 21, kept open for the whole session, carrying commands and replies; and a **data connection** on port 20, opened for each file transfer.
- **Client-server**: the client runs on the local host, the server on the remote host; the client initiates both connections.
- **Authentication**: USER and PASS; anonymous FTP for public files.
- **Transfer modes**: ASCII for text and binary (image) for programs, so line endings are not corrupted.
- **Resumable and abortable**: because control is separate from data, a transfer can be paused, resumed or aborted (ABOR) without losing the session.

Properties: reliable (runs over TCP), stateful (the server remembers the current directory and the login), efficient for bulk transfer, and secure only to the extent that the password is protected - plain FTP sends it in clear text, which is why FTPS and SFTP exist.

- **Advantages**: fast, efficient, resumable, and it works between different operating systems.
- **Limitations**: a file size limit of about 2 GB, no simultaneous transfer to multiple receivers, and not compatible with every system.

Q24. Explain the SMTP protocol and its role in Email.

- **SMTP** (Simple Mail Transfer Protocol) sets the rules for **sending** e-mail; it is the push protocol of the mail system.
- It is an application-layer protocol that uses **TCP port 25**, so it is connection-oriented and reliable.
- It is a **pure text protocol**, so it can carry text only; **MIME** extends it to images, audio, video and attachments.
- **Architecture**: user agent - mail queue - sender MTA - TCP port 25 - receiver MTA - mailbox - receiver user agent.
- **Role**: it moves a message from the sender's server to the recipient's server and between servers along the way; it never touches the recipient's mailbox directly - reading is done by POP3 or IMAP.

1. The user agent composes the message and hands it to the mail queue.
2. The sender MTA opens a TCP connection to port 25 of the receiving MTA.
3. The two MTAs exchange commands and replies; the message is transferred.
4. The receiving MTA stores it in the recipient's mailbox.
5. The recipient's user agent downloads it with POP3 (port 110) or IMAP (port 143).

Ports to remember: SMTP 25 push, POP3 110 pull, IMAP 143 pull; secure versions 587, 995 and 993.

Q25. Differentiate between Guided Media and Unguided Media. Give two examples of each type and briefly explain the characteristics of their signal transmission. List out four major comparisons between Coaxial Cable and Optical Fiber.

Point	Guided media	Unguided media
Conduit	A physical cable guides the signal	No cable - free space
Examples	Twisted pair, coaxial cable, optical fiber	Radio waves, microwaves, infrared
Signal	Voltage or light confined in the cable	Electromagnetic wave radiated through air
Direction	Point-to-point along the cable	Broadcast (radio) or beam (microwave)
Security	Confined, harder to tap	Radiates, anyone in range can receive
Impairment	Attenuation, crosstalk	Attenuation, interference, multipath

- **Twisted pair** - two insulated copper wires twisted so that induced noise cancels; signal is a differential voltage; cheap, short range.
- **Optical fiber** - light in a glass core confined by total internal reflection; huge bandwidth and immunity to EMI.
- **Radio waves** - omni-directional, penetrate walls, used for multicast; 3 kHz to 1 GHz.
- **Microwaves** - directional, line of sight, cannot penetrate walls; used for cellular, satellite and WLAN.

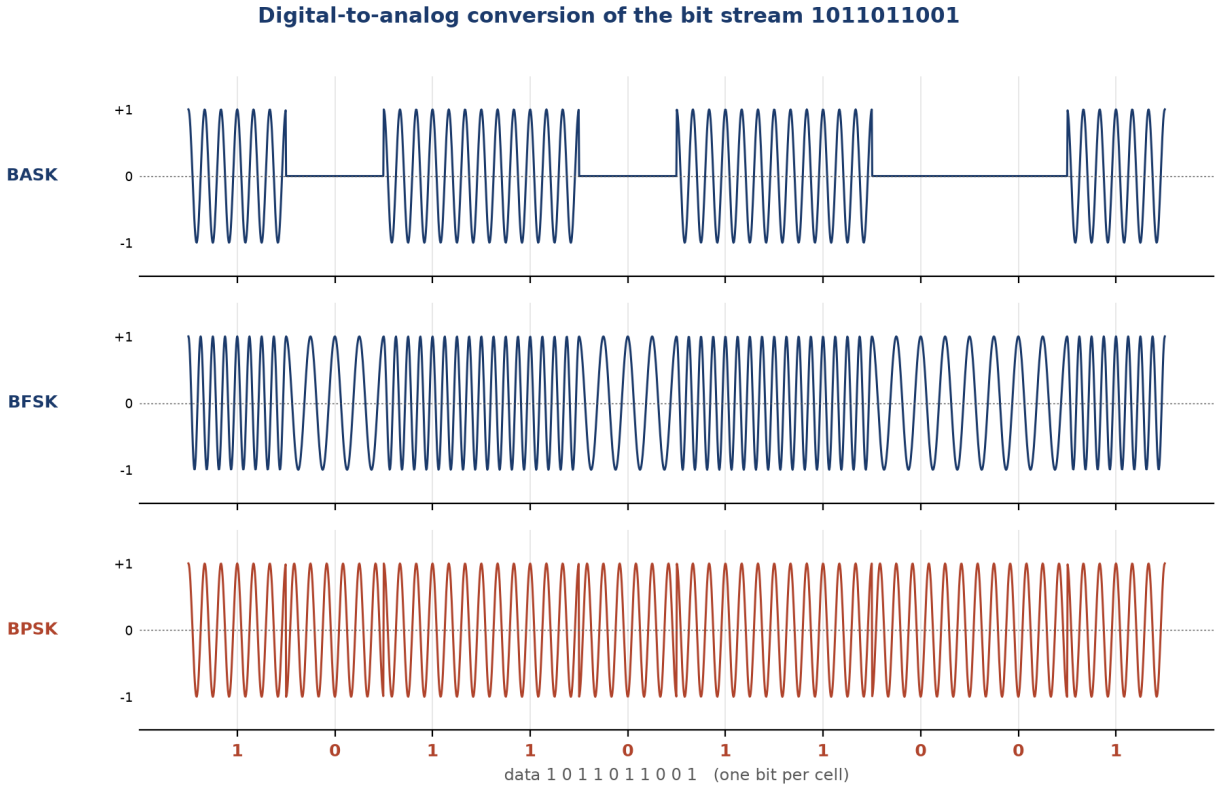
Four comparisons - coaxial cable versus optical fiber.

Point	Coaxial cable	Optical fiber
Bandwidth	Moderate - hundreds of MHz	Very high - Tbps
Attenuation and distance	High; repeaters every few km	Low; repeaters every tens of km
Interference	Shielded but still affected by EMI and crosstalk	Completely immune - the signal is light
Cost, weight and installation	Cheap and easy to splice, but heavy	Costlier, needs skilled splicing, but lighter and thinner

Memory hook - guided = **T-C-F**, unguided = **R-M-I**, going from wide coverage to short range.

Q26. Explain the concept of modulation of digital data. Discuss the different techniques used for digital-to-analog conversion, such as Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK), with suitable diagrams and examples.

Concept. Digital-to-analog conversion changes one characteristic of a carrier - C(t) = A cos(2 pi fc t + phi) - according to the digital data. Bit rate N and baud rate S are related by S = N x 1/r, and the minimum bandwidth is B = (1 + d)S for ASK and PSK.



1. **ASK** - amplitude switched: 1 sends the carrier, 0 sends zero voltage; generated by multiplying unipolar NRZ with the carrier. Example: an available band of 100 kHz from 200 to 300 kHz with d = 1 gives fc = 250 kHz, S = 50 kbaud and N = 50 kbps.
2. **FSK** - frequency switched: f1 = fc + delta-f for 1 and f2 = fc - delta-f for 0; generated with NRZ and a voltage-controlled oscillator. Constant amplitude, so it rejects amplitude noise, but it needs the widest band.
3. **PSK** - phase switched: in BPSK, 1 sends phase 0 and 0 sends phase 180; generated by multiplying polar NRZ with the carrier. Same bandwidth as ASK with the best noise immunity.
4. **Multilevel and QAM** - with L levels or L x M points each symbol carries log2(L x M) bits, so the bit rate rises without widening the band; 16-QAM carries 4 bits per symbol.

Comparison to write: ASK is simplest and weakest; FSK is robust but wide; PSK is the best compromise, and QAM gives the highest bits per symbol at the cost of a stricter SNR.

Q27. Explain the concept of modulation of analog signals. Discuss Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM). Compare these techniques based on bandwidth, signal quality, and applications.

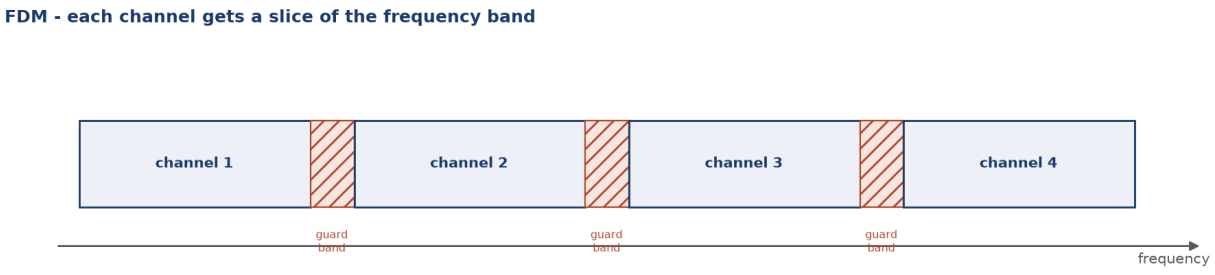
1. **AM** - the amplitude of the carrier follows the modulating signal, so the modulating signal becomes the envelope of the carrier. Bandwidth 2B.
2. **FM** - the frequency of the carrier follows the modulating signal; the amplitude stays constant. Bandwidth 2(1 + beta)B, where beta is the modulation index.
3. **PM** - the phase of the carrier follows the modulating signal; the phase change appears as a frequency change proportional to the derivative of the modulating signal. Bandwidth as for FM.

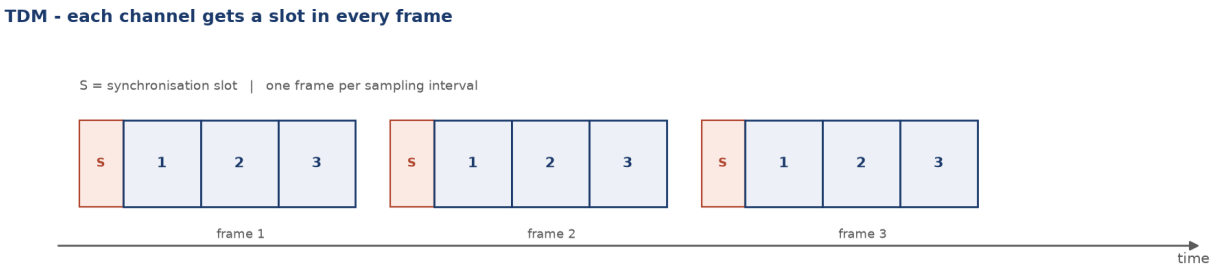
Point	AM	FM	PM
Parameter varied	Amplitude	Frequency	Phase
Bandwidth	2B	2(1 + beta)B	2(1 + beta)B
Signal quality	Poor - noise adds to amplitude	Good - constant envelope rejects amplitude noise	Good - same as FM
Hardware	Simple, cheap	More complex	More complex
Application	AM broadcast 535-1605 kHz, 10 kHz per station	FM broadcast 88-108 MHz, 200 kHz per station	Data modems, satellite and some mobile systems

Reading the table. AM is the cheapest in bandwidth and the weakest in quality: any noise on the amplitude goes straight into the audio. FM and PM pay (1 + beta) times the bandwidth for a constant envelope, which a limiter can clean. That is why music is broadcast on FM and speech on AM, and why PM is used inside digital schemes where the phase is easy to generate coherently.

Q28. What is multiplexing? Explain the different multiplexing techniques used in communication systems. Discuss FDM, WDM and TDM with suitable diagrams and applications.

Definition. Multiplexing allows several signals to share one link. The multiplexer combines n streams into one and the demultiplexer separates them at the far end. It is used whenever the link's bandwidth exceeds the needs of one signal.





- **FDM** - the band is divided into slices, one per signal, with guard bands between them; each signal is modulated onto its own carrier and all travel at once. Applications: AM and FM broadcast, cable TV, first-generation telephone carrier systems.
- **WDM** - FDM at optical frequencies; several wavelengths share one fiber, combined and split by a prism or grating. Applications: fiber backbones, undersea cables, where one fiber carries many 10 or 40 Gbps channels.
- **TDM** - time is divided into frames and slots; every channel uses the whole bandwidth but takes turns. Applications: the digital telephone hierarchy (DS-0 at 64 kbps up to DS-4), SONET/SDH, ISDN.

- **Synchronous TDM** wastes slots on idle streams; **statistical TDM** allocates slots on demand and needs an address per slot.
- **Numeric check**: five 100 kHz channels with 10 kHz guard bands need 5 x 100 + 4 x 10 = 540 kHz; 32 channels of 8 bits at 8 kHz need 2.048 Mbps.

One-liner: FDM divides the spectrum, TDM divides the clock, WDM divides the colour.

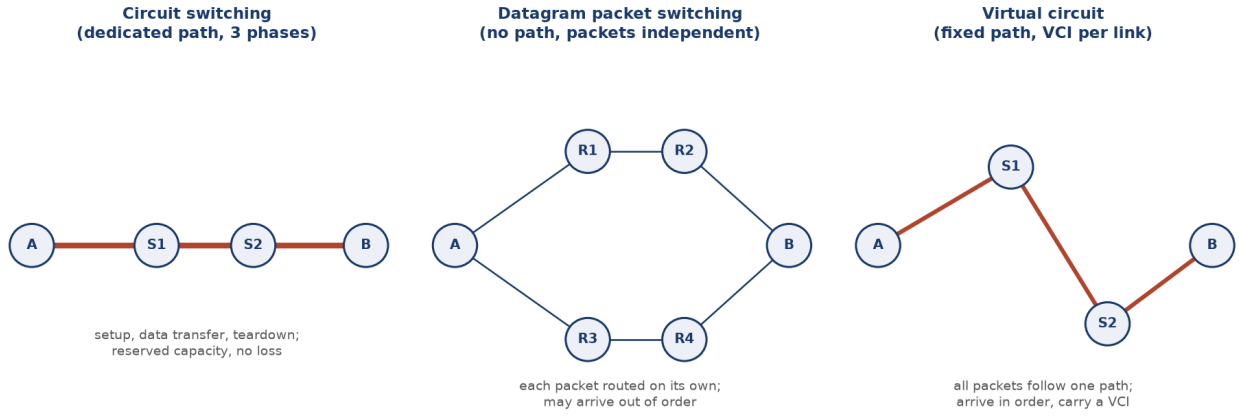
Q29. Explain guided and unguided transmission media in detail. Compare twisted-pair cable, coaxial cable, and optical fiber under guided media, and radio waves, microwaves, and infrared under unguided media. Discuss their advantages, disadvantages, and applications.

Medium	Construction	Advantages	Disadvantages	Application
Twisted pair (UTP/STP)	Two insulated copper wires twisted in a helix	Cheap, easy to install, flexible	Low bandwidth, short range, crosstalk	Telephone loop, Ethernet, LAN
Coaxial	Core, dielectric, braided shield, jacket	Better shielding and bandwidth than twisted pair, longer runs	Bulky, expensive to install, still affected by EMI	Cable TV, broadband Internet, older Ethernet
Optical fiber	Glass or plastic core with cladding, light source	Tbps bandwidth, low attenuation, immune to EMI, secure	Costly, skilled splicing, fragile, one-way per strand	Backbones, undersea cables, FTTH

Medium	Propagation	Advantages	Disadvantages	Application
Radio waves	Omni-directional, penetrates walls	Wide coverage, no line of sight needed, cheap	Low bandwidth, interference, poor security	AM and FM radio, TV, paging, cordless phones
Microwaves	Directional, line of sight	High bandwidth, long distance with a dish	Cannot penetrate walls, needs alignment and towers, rain fade	Cellular, satellite, WLAN, point-to-point links
Infrared	Line of sight, short range, cannot pass walls	Cheap, no interference between rooms, secure in a room	Very short range, blocked by any obstacle	Remote controls, short-range device links

How to answer in the exam: one table for guided, one for unguided, then one line - guided media are for capacity and security, unguided media are for mobility and reach.

Q30. Explain circuit switching and packet switching in detail. Describe their working principles and compare them in terms of connection establishment, resource utilization, delay, reliability, and applications. Illustrate the operation of both switching techniques with suitable diagrams.



1. **Circuit switching** - physical layer. Setup: the source requests a connection that every switch must accept. Data transfer: a continuous stream on a reserved path, the link divided into channels by FDM or TDM. Teardown: the circuit is released. Delay after setup is only propagation, but capacity is wasted while the circuit is silent.

2. **Packet switching (datagram)** - network layer. The message is divided into packets with the full destination address; each packet is routed independently, first-come-first-served, and may take a different route, arrive out of order or be dropped.
3. **Virtual circuit** - a cross between the two. A path is set up before transfer, so packets arrive in order, but no capacity is reserved; each packet carries a VCI that the switches use for table lookup.

Criterion	Circuit switching	Packet switching
Connection establishment	Required - setup phase with reservation	Not required (datagram); setup only for a virtual circuit
Resource utilisation	Poor - reserved capacity is wasted when idle	Good - links are shared statistically
Delay	Setup delay, then constant and small	Queueing delay per packet, variable and load dependent
Reliability	Order guaranteed, no loss, but a link failure drops the call	Robust - packets reroute around a failure, but may be lost or reordered
Applications	Telephone network, leased lines	The Internet, e-mail, web, file transfer

Best closing line: circuit switching suits constant-rate traffic that needs guaranteed delay; packet switching suits bursty traffic that values link efficiency over delay guarantees.

Q31. Explain the error detection process in detail. Discuss parity checking, checksum, and CRC techniques with suitable examples. Compare their effectiveness in detecting transmission errors.

Process. The sender adds redundant bits derived from the data; the receiver recomputes them and compares. If the check fails the frame is discarded and, at the data link layer, retransmitted. Detection needs a minimum Hamming distance of s + 1 to catch s errors.

1. **Single parity check** - one extra bit makes the total number of 1s even (or odd). Example: data 1011001 has four 1s, so even parity appends 0 and sends 10110010; a single flipped bit makes the count odd and is caught. It detects any odd number of errors and misses every even number.
2. **Two-dimensional parity** - parity for every row and every column, which catches most bursts and can even locate a single-bit error.
3. **Checksum** - the message is divided into m-bit units, added with one's complement wraparound, and the complement of the sum is sent. Example with 4-bit units 7, 11, 12, 0, 6: the sum wraps to 6, so the checksum is 9 and the receiver's addition gives all 1s. Used in IP, TCP and UDP.
4. **CRC** - the dataword is extended with n-k zeros and divided modulo-2 by a generator polynomial; the remainder is the CRC and is appended. At the receiver the whole codeword is divided again - a non-zero remainder means an error. It is cheap in hardware (shift registers) and very strong against bursts.

Technique	Redundancy	Detects	Cost	Used in
Single parity	1 bit	Odd number of errors	Lowest	Simple links
Two-dimensional parity	Row and column bits	Most bursts	Low	Older protocols
Checksum	m bits	Most errors, but weak against some reorderings	Low in software	IP, TCP, UDP
CRC	16 or 32 bits	All single, double and odd errors, most bursts	Low in hardware	Ethernet, HDLC, Wi-Fi

Effectiveness ranking: CRC is the strongest per bit of redundancy and is used wherever hardware is available; checksum is the software-friendly choice; parity is the weakest and is used only where nothing better is affordable.

Q32. Explain the concept of error correction in computer networks. Describe how Hamming Code can be used to detect and correct errors. Demonstrate the error-correction process with a suitable example.

Concept. Detection only tells the receiver that something is wrong; correction needs enough redundancy to locate the error. Forward error correction adds redundancy so the receiver can repair the data without a retransmission - essential on a satellite link where a retransmission would take too long. Correction needs a minimum Hamming distance of 2t + 1 to correct t errors.

1. **Positions**: r parity bits occupy the positions that are powers of 2 (1, 2, 4, 8 ...), and the data fill the rest. The condition is 2^r >= m + r + 1; for m = 4, r = 3 gives the (7,4) code.
2. **Parity equations**: P1 checks positions 1, 3, 5, 7; P2 checks 2, 3, 6, 7; P4 checks 4, 5, 6, 7.
3. **Encoding example**, data 1011 in positions 7, 6, 5, 3: P1 = 1 XOR 1 XOR 1 = 1, P2 = 1 XOR 0 XOR 1 = 0, P4 = 1 XOR 0 XOR 1 = 0. Codeword = **1010101**.
4. **Error case**: suppose bit 4 flips, so 1011101 is received. Recompute: S1 = 1 XOR 1 XOR 1 XOR 1 = 0; S2 = 0 XOR 1 XOR 0 XOR 1 = 0; S4 = 1 XOR 1 XOR 0 XOR 1 = 1.
5. **Syndrome** S4 S2 S1 = **100**, which is binary 4 - the syndrome *is* the position of the error. Flip bit 4 back and the data 1011 is recovered.

Two things to say for the marks: the syndrome gives the position in binary, and the code corrects one error or detects two (with an extra overall parity bit, it can do both).

Q33. A sender wants to transmit the dataword 1011011 using CRC with the generator polynomial G(x) = x^3 + x + 1. (a) Determine the transmitted codeword. (b) Assume that one bit is altered during transmission. Explain how the receiver detects the error using CRC.

1. G(x) = x^3 + x + 1 is the bit pattern **1011**, so n - k = 3 and three zeros are appended.
2. Augmented dataword: 1011011**000** = 1011011000.
3. Step 1: 1011 XOR 1011 = 0000, bring down 1 and 1 - the leading bit is 0, so skip until 1100.
4. Step 2: 1100 XOR 1011 = **0111**.
5. Step 3: 1110 XOR 1011 = **0101**.
6. The last 3 bits are the remainder: **CRC = 101**.
7. **Transmitted codeword = 1011011 101 = 1011011101**.

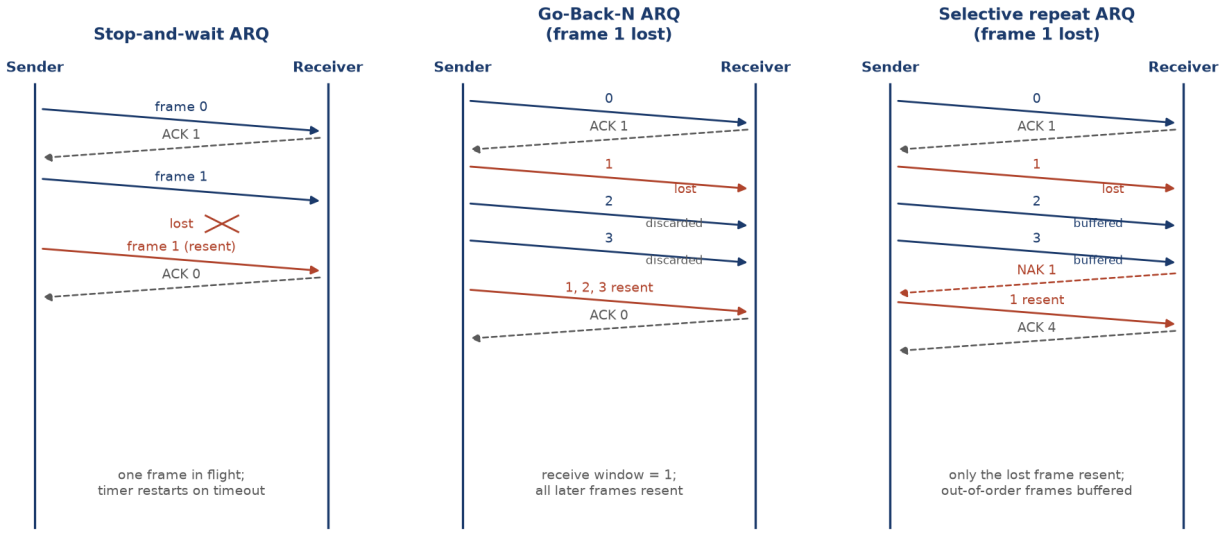
(b) Detection at the receiver. The receiver divides the whole received codeword by the same generator 1011 and looks at the remainder:

- If the remainder is **000**, no error is assumed and the dataword 1011011 is extracted.
- If the remainder is **non-zero**, an error has occurred and the frame is discarded, which at the data link layer triggers a retransmission.
- Example: suppose bit 4 is flipped, so 1010011101 arrives. Dividing by 1011 leaves remainder **101**, not 000, so the error is detected.
- CRC cannot correct the error - it only detects it; the correction comes from ARQ, the retransmission of the frame.

Why CRC is strong: a generator of degree r catches all burst errors of length r or less, and with a 32-bit generator the probability of an undetected error is about 1 in 4 billion.

Q34. Explain the Stop-and-Wait ARQ protocol in detail. Describe its working using a suitable example showing ACK, timeout, retransmission, and lost/damaged frames. Discuss its advantages and limitations.

1. The sender sends **one frame**, keeps a copy, starts a timer and waits.
2. The receiver checks the frame and replies with an ACK that announces the number of the **next frame expected**.
3. If the ACK arrives before the timer expires, the sender sends the next frame; if the timer expires, it retransmits the same frame.
4. Sequence numbers are **modulo 2** (0, 1, 0, 1) because only two frames are ever outstanding.



- **Case 1 - normal**: frame 0 is sent and acknowledged, then frame 1.
- **Case 2 - frame lost**: the timer expires, frame 0 is resent and this time acknowledged.
- **Case 3 - ACK lost**: the sender cannot tell whether the frame or the ACK was lost, so it resends the frame; the receiver sees a duplicate sequence number, discards it and re-acknowledges.
- **Case 4 - frame damaged**: the receiver detects the error by CRC, discards the frame and sends a NAK (or simply stays silent), so the sender retransmits on timeout.
- **Advantages**: simplest possible protocol, needs only one frame of buffer at each end, and is easy to implement and to prove correct.
- **Limitations**: only one frame in flight, so the sender is idle for a round-trip time - utilisation is about 1/(1 + 2a) where a is the propagation-to-transmission-time ratio, which is very poor on a long fast link; a lost ACK causes a duplicate transmission; and it wastes bandwidth on every recovery.

When it is fine: a short link with a low data rate, such as a serial line or a sensor bus, where the propagation delay is negligible against the frame time.

Q35. Explain HDLC (High-Level Data Link Control) in detail. Discuss its frame format, frame types (I-frame, S-frame, and U-frame), control field, flow control, and error control mechanisms.

- **HDLC** is a bit-oriented protocol for point-to-point and multipoint links, providing framing, flow control and error control.
- **Frame format**: flag (01111110) - address - control - information - FCS - flag. The flag delimits the frame and bit stuffing keeps it unique.
- **I-frame** - carries user data with N(S), the send sequence number, and N(R), a piggybacked acknowledgement.
- **S-frame** - control only: RR (receive ready), RNR (receive not ready), REJ (reject, go-back-N), SREJ (selective reject).
- **U-frame** - link management: SNRM/SNRME (set normal response mode), SABM/SABME (set asynchronous balanced mode), DISC (disconnect), UA (unnumbered acknowledgement), FRMR (frame reject).
- **Control field** - the first one or two bits give the frame type (0 = I, 10 = S, 11 = U); for I-frames it then carries N(S) and N(R), for S-frames the S-function and N(R), for U-frames the modifier bits. The P/F bit is the poll bit on a command and the final bit on a response.

Flow control works by sliding windows: the receiver advertises its readiness with RR, withdraws it with RNR, and the sender may have at most 2^m - 1 frames outstanding (128 in the 3-bit default). **Error control** works by ARQ: the FCS detects the error, REJ asks for a go-back-N retransmission, SREJ asks for one frame, and a timer at the sender covers a lost frame or a lost acknowledgement.

Exam line: HDLC's control field is the whole protocol in miniature - it carries the frame type, the sequence number, the acknowledgement and the poll/final bit.

Q36. Explain CSMA/CD and CSMA/CA in detail. Compare their working mechanisms, collision detection/avoidance techniques, applications, and limitations.

1. **CSMA/CD**: sense the medium. If idle, transmit; if busy, wait. Keep sensing **while** transmitting; on detecting a collision send a 32-bit jam signal, abort the frame, wait a random time from 0 to 2^k - 1 slot times (binary exponential backoff) and retransmit.
2. **CSMA/CA**: sense the medium. If idle, do not transmit at once - wait an **interframe space** (SIFS for an ACK, longer DIFS for data). Then choose a random number of slots from the **contention window** and count down while the medium stays idle. Transmit, and expect an **acknowledgement**; if none arrives, back off and retry.

Point	CSMA/CD	CSMA/CA
Collision handling	Detect during transmission	Avoid before transmission
Why	A wired station can hear the medium while sending	A radio cannot hear while sending, and hidden terminals exist
Mechanisms	Sensing, jam signal, backoff	IFS, contention window, ACK, optional RTS/CTS
Overhead	Low	High - waiting and ACKs
Efficiency	High on a wired bus	Lower, but the only practical choice on radio
Application	Traditional Ethernet (802.3)	Wireless LAN (802.11)

Limitations to mention: CD cannot be used on radio and degrades at high load; CA wastes capacity on waiting and ACKs and suffers from the hidden-terminal problem, which RTS/CTS only partly solves.

Q37. Explain Controlled Access techniques used in multiple-access networks. Discuss Reservation, Polling, and Token Passing with suitable examples. Compare their advantages and limitations.

- **Reservation** - time is divided into an interval with a reservation frame followed by data frames. The reservation frame has N mini slots, one per station; a station with data marks its own mini slot, and only those stations may send in the following data period. Example: a satellite channel with 100 earth stations, each with a mini slot.
- **Polling** - one station is the primary and the rest are secondaries. The primary sends a **select** (SEL) frame to give a station the right to receive, or a **poll** frame to ask it to send, in round-robin order. Example: a mainframe polling terminals, or Bluetooth's master polling its slaves.
- **Token passing** - stations form a logical ring and a special frame called the token circulates. A station may transmit only while it holds the token, then passes it on. Example: token ring and FDDI.

Point	Reservation	Polling	Token passing
Who decides	Pre-booked mini slots	The primary station	Whoever holds the token
Overhead	One reservation frame per interval	A poll frame per station per round	The token itself, plus ring latency
Delay	Predictable	Predictable but grows with the number of stations	Predictable, bounded by the token rotation time
Weakness	Idle slots wasted; needs synchronisation	Primary is a single point of failure	Lost token stops the network; needs token recovery

Why controlled access at all: it removes collisions, so delay is bounded - which is what real-time and industrial traffic needs, and what random access cannot promise at high load.

Q38. Explain the Client-Server Model in detail. Describe the roles of the client and server, request-response communication, server processes, and different types of servers with suitable real-life examples.

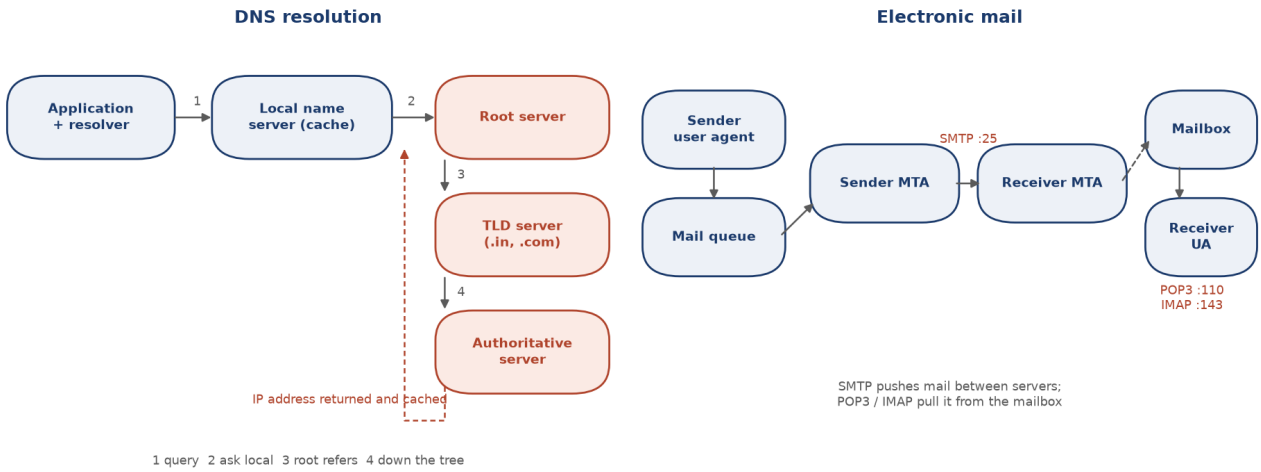
- **Client** - a program on the local machine that requests a service; **server** - a program on a remote machine that provides it. One server serves many clients at once.
- **Roles**: the client initiates, formats the request and interprets the reply; the server waits for requests, performs the work and returns the result. The roles are asymmetric - the server is passive, the client active.
- **Request-response**: the client sends a request to a well-known port; the server processes it and returns a response. HTTP's GET and 200 OK, and DNS's query and answer, are the standard examples.
- **Socket interface**: both sides reach the network through a socket, identified by the pair (IP address, port), so a connection is identified by four values - source IP and port, destination IP and port.
- **Server processes**: an **iterative** server handles one request at a time; a **concurrent** server forks a process or thread per request, or uses a pool of workers.
- **Types of servers**: web server (Apache, Nginx), mail server (SMTP and POP3/IMAP), file server (FTP, NFS), DNS server, database server, print server, proxy server.

Advantage	Disadvantage
Centralised backup and administration	Traffic congestion at the server
Central security control	Single point of failure
Better performance and easy scalability	Data may be modified in transit; the server must be trusted

Real-life example: a college web portal - the browser is the client, the web server holds the pages, the database server holds the marks, and the DNS server turns the portal's name into an address.

Q39. Explain the Domain Name System (DNS) in detail. Discuss the need for DNS, hierarchical naming structure, domain names, DNS servers, and the process of converting a domain name into an IP address.

- **Need**: routers need numeric IP addresses, people remember names. DNS is the distributed database that maps one to the other, and back.
- **Hierarchical naming**: the name space is a tree with the root at the top; labels are separated by dots and read from the specific to the general - www.cgu-odisha.ac.in is host www, domain cgu-odisha, then the second-level ac, then the country domain in.
- **Sections**: generic domains (.com, .org, .edu, .gov, .net, .mil, .int), country domains (.in, .us, .uk) and the inverse domain (in-addr.arpa) for reverse lookup.
- **Servers**: root, primary (authoritative), secondary and local (caching).
- **Properties**: client-server, uses UDP port 53, hierarchical, and caches answers for their TTL.



1. The application calls the **resolver** with a name query.
2. The resolver checks its cache; on a hit it returns the address at once.

3. On a miss it queries the **local name server**, which checks its cache and zone file.
4. If that server cannot answer, it asks the **root server**, which refers it to the top-level domain server, which refers it to the **authoritative server** for the domain.
5. The answer travels back and every server caches it for its TTL.
6. The resolver returns the IP address and the application opens a TCP connection to it.

Recursive versus iterative: in recursive resolution the server does all the work and returns the final answer; in iterative resolution it returns a referral and the client asks the next server itself.

Q40. Explain the Simple Mail Transfer Protocol (SMTP) in detail. Describe its purpose, client-server communication, commands, response codes, and the process of sending an email.

- **Purpose**: to transfer mail reliably from the sender's server to the recipient's server, and between servers along the way. It is a push, text-based, connection-oriented protocol on TCP port 25.
- **Client-server**: the sending MTA is the client, the receiving MTA is the server; the client opens the connection and issues commands, the server answers with a code and a phrase.
- **Commands**: HELO or EHLO (identify and open the session), MAIL FROM (sender address), RCPT TO (recipient), DATA (start the message body, ended by a line containing only a full stop), RSET (abort the transaction), VRFY (verify a user), NOOP, QUIT.
- **Response codes**: 220 service ready, 250 requested action completed, 354 start mail input, 421 service not available, 450/451/452 temporary failures, 500 syntax error, 550 mailbox unavailable, 552/554 transaction failed - the first digit says class (2 success, 3 intermediate, 4 transient, 5 permanent).

1. The user agent composes the message and puts it in the mail queue.
2. The sender MTA connects to port 25 of the receiving MTA and they exchange EHLO, MAIL FROM and RCPT TO.
3. DATA transfers the headers and body; the message ends with a full stop on its own line.
4. The receiving MTA stores the message in the recipient's mailbox and acknowledges.
5. The recipient's user agent retrieves it with POP3 (110) or IMAP (143); MIME decoding restores any attachments.

Why a separate read protocol: SMTP only pushes; pulling mail from a mailbox needs POP3 or IMAP.

Q41. Explain FTP (File Transfer Protocol) in detail. Discuss its architecture, control connection, data connection, authentication, file transfer process, and common FTP operations.

1. **Architecture**: client-server, with two processes on each side - a control process and a data transfer process. The client runs on the local host and initiates both connections.
2. **Control connection**: TCP port 21, opened first, kept open for the whole session, carries commands and replies in NVT ASCII.
3. **Data connection**: TCP port 20, opened for each file transfer and closed when that file is complete. In **active mode** the server opens it to the client; in **passive mode** the client opens it, which is what works through firewalls.
4. **Authentication**: USER and PASS over the control connection, or anonymous FTP with an e-mail address as the password for public files.
5. **Transfer process**: the client sends RETR or STOR over the control connection; the data connection is opened, the file is copied, the data connection closes, and the control connection reports completion. ASCII mode is used for text and binary mode for programs, so line endings are not corrupted.
6. **Common operations**: RETR (download), STOR (upload), LIST (directory listing), NLST (names only), CWD (change directory), MKD and RMD, DELE, RNFR/RNTO (rename), ABOR (abort), QUIT.

- **Advantages**: fast, resumable, works across operating systems, and the two-connection design lets a transfer be monitored and aborted.
- **Limitations**: about a 2 GB file size limit, no simultaneous transfer to multiple receivers, not compatible with every system, and plain FTP sends the password in clear text.

One-line answer: FTP separates *talking about the file* (port 21) from *carrying the file* (port 20) - that separation is the design idea examiners look for.

Q42. A company has been assigned the network 192.168.50.0/24 and wants to create 32 equal-sized sub-networks. Determine (i) the number of subnet bits, (ii) the new subnet mask, (iii) the IP addresses per subnet, (iv) the usable hosts per subnet, (v) the network ID and broadcast address of the 1st, 10th, 20th and 32nd subnets, and (vi) their usable ranges.

1. **Subnet bits**: 2^5 = 32, so **5 bits** are borrowed from the host part.
2. **New subnet mask**: /24 + 5 = **/29**, that is 255.255.255.248 (248 = 11111000).
3. **IP addresses per subnet**: 2^(32-29) = 2^3 = **8**.
4. **Usable hosts per subnet**: 8 - 2 = **6**.
5. **Block size** = 8 in the fourth octet, so the subnets step 0, 8, 16, 24 ... 248; the k-th subnet starts at (k - 1) x 8.

Subnet	Network ID	Broadcast	Usable range
1st	192.168.50.0	192.168.50.7	192.168.50.1 - 192.168.50.6
10th	192.168.50.72	192.168.50.79	192.168.50.73 - 192.168.50.78
20th	192.168.50.152	192.168.50.159	192.168.50.153 - 192.168.50.158
32nd	192.168.50.248	192.168.50.255	192.168.50.249 - 192.168.50.254

Check: the 32nd subnet is the last one, and its broadcast is 192.168.50.255 - the broadcast of the original /24, which is the arithmetic proof that the whole block has been used.

Q43. For the given IP address 192.168.20.0, the network has been divided into 4 sub-networks. Determine (i) the network ID, (ii) the default subnet mask, (iii) the new subnet mask, (iv) the IP addresses per subnet, (v) the usable hosts per subnet, (vi) the broadcast address of each subnet and (vii) the range of addresses in each subnet.

1. **Network ID**: 192.168.20.0. First byte 192 lies in 192-223, so it is a **class C** address.
2. **Default subnet mask**: 255.255.255.0, that is /24.
3. **Subnet bits**: 4 subnets need log2 4 = **2 bits**, so the new prefix is /26 and the **new subnet mask** is **255.255.255.192** (192 = 11000000).
4. **IP addresses per subnet**: 2^(32-26) = 2^6 = **64**.
5. **Usable hosts per subnet**: 64 - 2 = **62**.
6. **Block size** = 64, so the subnets are .0, .64, .128 and .192.

Subnet	Network ID	Broadcast	Usable range
1	192.168.20.0	192.168.20.63	192.168.20.1 - 192.168.20.62
2	192.168.20.64	192.168.20.127	192.168.20.65 - 192.168.20.126
3	192.168.20.128	192.168.20.191	192.168.20.129 - 192.168.20.190
4	192.168.20.192	192.168.20.255	192.168.20.193 - 192.168.20.254

Total check: 4 subnets x 64 addresses = 256, exactly the size of the original /24 block - which is how you know the arithmetic is right.

Q44. Discuss process-to-process delivery in the transport layer. Compare UDP and TCP protocols based on reliability, connection establishment, flow control, and applications.

- **Process-to-process delivery** means handing the data to the correct application on the destination host, not merely to the host. It is the job of the transport layer, and it is what distinguishes it from the network layer's host-to-host delivery.
- **Port numbers** make it possible: a 16-bit number identifies each process, so the pair (IP address, port) - the socket - identifies each end, and four values identify a connection.
- **Multiplexing and demultiplexing**: the sender's transport layer collects data from several processes and adds port numbers; the receiver's transport layer reads the port number and delivers each segment to the right process.
- **Port ranges**: 0-1023 well known (FTP 21, SMTP 25, DNS 53, HTTP 80), 1024-49151 registered, 49152-65535 dynamic.

Criterion	UDP	TCP
Reliability	Unreliable - no ACK, no retransmission, data may be lost	Reliable - ACK, retransmission, checksum, ordering
Connection establishment	None - each datagram is independent	Three-way handshake (SYN, SYN+ACK, ACK) before data
Flow control	None; the sender may overrun the receiver	Sliding window advertised by the receiver, plus congestion control
Header and overhead	8 bytes	20 to 60 bytes
Ordering and duplication	No ordering, duplicates possible	Ordered byte stream, duplicates discarded
Applications	DNS, DHCP, SNMP, streaming, VoIP, online games	HTTP, HTTPS, FTP, SMTP, Telnet

How to choose: use TCP when every byte must arrive, in order (a file, a web page, an e-mail); use UDP when late data are useless anyway and a retransmission would only add delay (voice, video, a DNS query).